

Interactive graph-based activity for divisibility

Application example

THE NUMBER 895958 IS DIVISIBLE BY 7

This document presents, through a practical example, how to use the interactive graph-based activity to study the divisibility of a number. The selected example is the number 895958 and the modulus 7. The objective is to verify, by constructing a divisibility graph, that $895958 \pmod{7} = 0$ and, therefore, that 895958 is divisible by 7.

The activity allows this process to be carried out visually and interactively. Students must first construct the transition graph corresponding to the chosen modulus, then perform the walk (or sequence of states) associated with the digits of the number and finally determine the remainder and decide whether the number is divisible.

The procedure is developed in three phases: (1) construction and verification of the graph; (2) construction and verification of the walk; and (3) determination of the remainder and divisibility. In this way, the user is not given the result directly but must progressively construct the procedure that leads to it.

Throughout the activity, the application guides the user and allows them to check whether each step they perform is correct.



© 2026 Daniel Martín-Cudero · CC BY 4.0
Creative Commons Attribution 4.0 International

USER MANUAL

STEP 1. CONSTRUCTION AND VERIFICATION OF THE GRAPH

First, the modulus to be studied is entered into the application. The resource allows users to select moduli from 2 to 20. In this case, 7 is entered. When the “Create graph” button is pressed, the application automatically generates the seven states of the graph and arranges them around a circle. The states displayed are: 0, 1, 2, 3, 4, 5, and 6. These states represent the seven possible remainders when dividing by 7. Figure 1 shows the interface after entering modulus 7, with the seven states arranged in a circle.

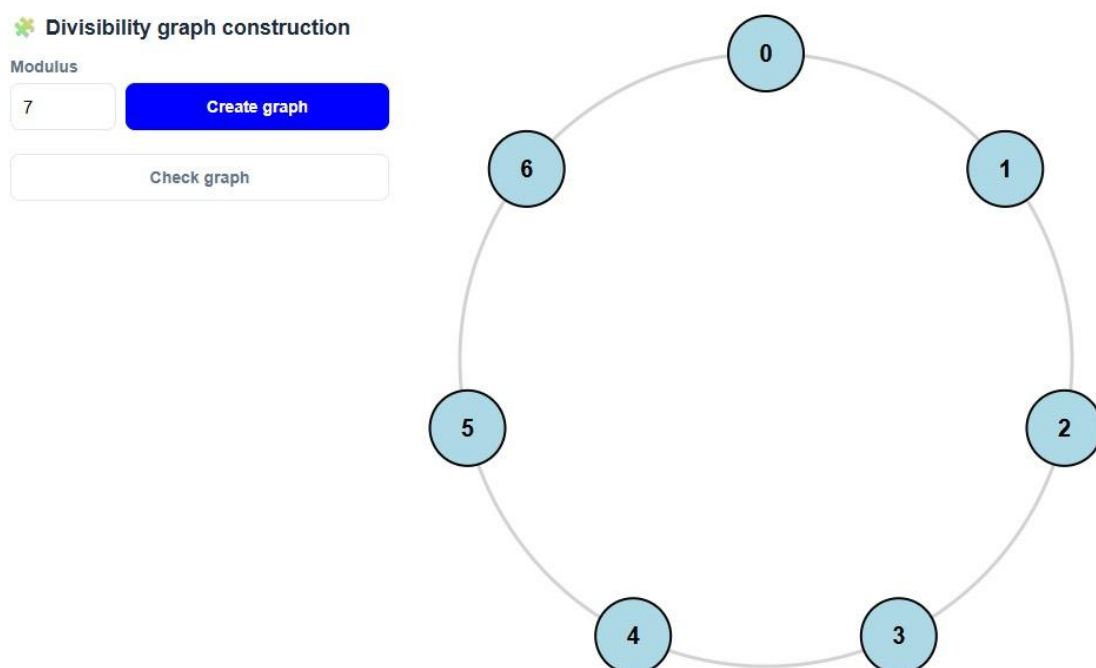


Figure 1. Application interface after entering modulus 7 and generating the graph.

Once the graph has been generated, students must manually construct the transitions between the states of the graph. The rule to be used is:

$$r \rightarrow (10 \cdot r) \bmod 7,$$

where ‘r’ is the source state. That is, for each state in the graph, the corresponding destination state must be calculated. The seven transitions are shown in Table 1.

Table 1. Transitions between the states of the divisibility graph modulo 7.

Source state	Destination state	Operation
0	0	$(10 \cdot 0) \bmod 7 = 0$
1	3	$(10 \cdot 1) \bmod 7 = 3$
2	6	$(10 \cdot 2) \bmod 7 = 6$
3	2	$(10 \cdot 3) \bmod 7 = 2$

Source state	Destination state	Operation
4	5	$(10 \cdot 4) \bmod 7 = 5$
5	1	$(10 \cdot 5) \bmod 7 = 1$
6	4	$(10 \cdot 6) \bmod 7 = 4$

To enter each transition into the application, the source node is selected first, followed by the destination node. The application automatically draws a gray arrow between the two nodes. Figure 2 shows the graph after all seven transitions between states have been entered.

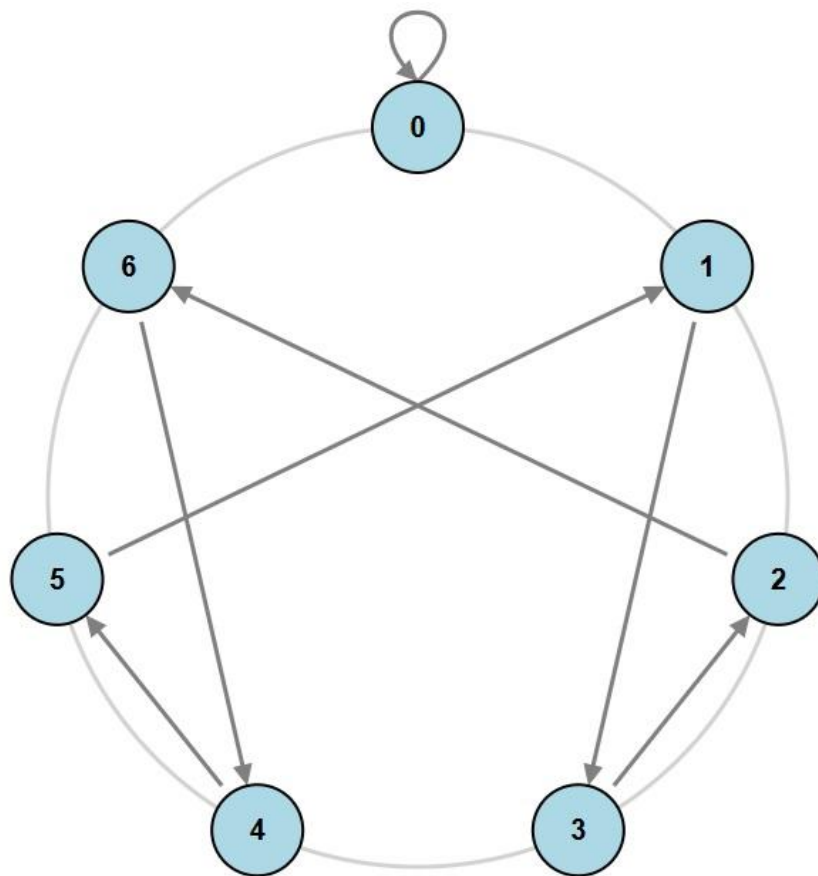


Figure 2. Graph with all transitions entered.

Once all the transitions have been constructed, the “Check graph” button is pressed. If the transitions are correct, the application informs the user that the graph has been built correctly. If they are not correct, the application indicates that the graph is incorrect. In this case, the user must press the “Try it again” button to start over.

 The graph has been built correctly. You can now start the traversal. Enter the number and press the “Start traversal” button.

 The graph is not correct

Try it again

Figure 3. Graph verification message.

STEP 2. CONSTRUCTION AND VERIFICATION OF THE WALK

Once the graph has been validated, the walk corresponding to the number to be checked can be performed. The number 895958 is entered and the “Start traversal” button is pressed.

The walk of the graph always begins at state $r_0 = 0$ (no digit of the number has been processed yet). From this point onward, the user must select the different states through which the graph passes as it successively processes the digits 8, 9, 5, 9, 5, and 8. The rule that determines each new state is:

$$r_0 = 0; \quad r_i = (10 \cdot r_{i-1} + d) \bmod 7 \quad (i = 1, 2, \dots, m-1),$$

where ‘d’ is the digit being processed. The steps for constructing the sequence of states are described below:

1. The source node is selected; in this case, node 0. According to the graph, the transition from state 0 leads back to state 0. Starting from state 0, eight positions are counted clockwise, one for each unit of the digit being processed. The traversal ends at state 1. Therefore, node 1 is selected to continue the walk. The application automatically draws a red arrow from node 0 to node 1.
2. The current state is 1. According to the graph, state 1 leads to state 3. Starting from state 3, nine positions are counted clockwise. The traversal ends at state 5. Therefore, node 5 is selected. The application draws a red arrow from node 1 to node 5.
3. The current state is 5. According to the graph, state 5 leads to state 1. Starting from state 1, five positions are counted clockwise. The traversal ends at state 6. Therefore, node 6 is selected. The application draws a red arrow from node 5 to node 6.
4. The current state is 6. According to the graph, state 6 leads to state 4. Starting from state 4, nine positions are counted clockwise. The traversal ends again at state 6. Therefore, node 6 is selected again. Since the traversal starts and ends at the same node, the application draws a red self-loop from node 6 to itself.
5. The current state is 6. According to the graph, state 6 leads to state 4. Starting from state 4, five positions are counted clockwise. The traversal ends at state 2. Therefore, node 2 is selected. The application draws a red arrow from node 6 to node 2.
6. The current state is 2. According to the graph, state 2 leads to state 6. Starting from state 6, eight positions are counted clockwise. The traversal ends at state 0. Therefore, node 0 is selected. The application draws a red arrow from node 2 to node 0.

Table 2. Calculation of the states traversed when processing the digits of the number 895958.

Source state	Destination state
$r_0 = 0$	$r_1 = (10 \cdot r_0 + 8) \bmod 7 = (10 \cdot 0 + 8) \bmod 7 = 1$
$r_1 = 1$	$r_2 = (10 \cdot r_1 + 9) \bmod 7 = (10 \cdot 1 + 9) \bmod 7 = 5$
$r_2 = 5$	$r_3 = (10 \cdot r_2 + 5) \bmod 7 = (10 \cdot 5 + 5) \bmod 7 = 6$
$r_3 = 6$	$r_4 = (10 \cdot r_3 + 9) \bmod 7 = (10 \cdot 6 + 9) \bmod 7 = 6$
$r_4 = 6$	$r_5 = (10 \cdot r_4 + 5) \bmod 7 = (10 \cdot 6 + 5) \bmod 7 = 2$
$r_5 = 2$	$r_6 = (10 \cdot r_5 + 8) \bmod 7 = (10 \cdot 2 + 8) \bmod 7 = 0$

The sequence of states will be: $0 \rightarrow 1 \rightarrow 5 \rightarrow 6 \rightarrow 6 \rightarrow 2 \rightarrow 0$. The application will also display this sequence as text while the walk is being performed. This allows the user to check at all times that the selected states are correct. If an error is made at any point, the “Start traversal” button can be pressed again to restart the walk from the beginning. Figure 4 shows the divisibility graph after entering the sequence of states generated by the number.

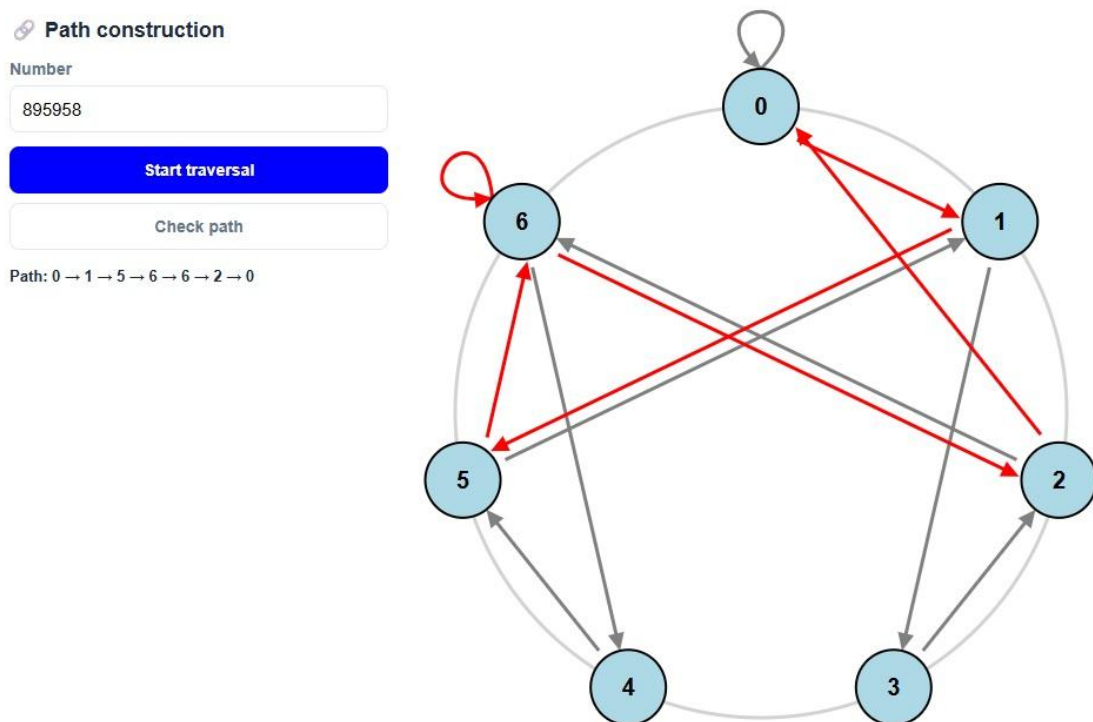


Figure 4. Graph with the complete walk corresponding to the number 895958.

Once the walk has been completed, the “Check walk” button must be pressed. If the walk is correct, the application will display the message “Correct walk”. The final phase of the activity will also be enabled, and the last state of the walk will be highlighted in green (see Figure 5). If the walk is incorrect, the application will display a message indicating the error and allow the user to try again.

Path construction

Number

895958

Start traversal

Check path

Path: 0 → 1 → 5 → 6 → 6 → 2 → 0

✓ Correct path

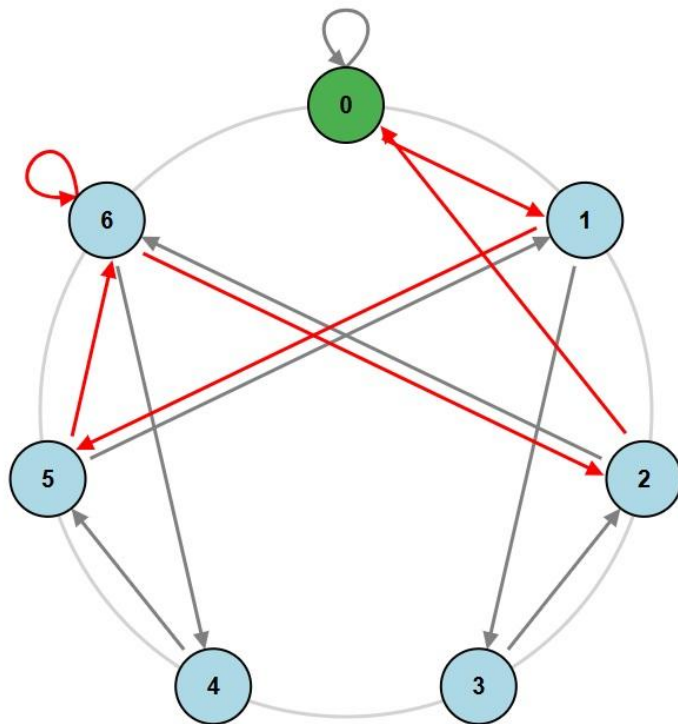


Figure 5. Correct walk and final state highlighted in green.

STEP 3. ENTERING THE REMAINDER AND CHECKING DIVISIBILITY

The final state of the walk directly represents the remainder when the number is divided by the modulus. In this case, the walk ends at state 0. Therefore, $895958 \pmod{7} = 0$, that is, the remainder when 895958 is divided by 7 is 0. Consequently, the division is exact, and the number 895958 is divisible by 7.

To complete the final step, the user must enter the obtained remainder into the application and indicate whether or not the number is divisible. The “Check final result” button must then be pressed. If the information entered is correct, the application will indicate that the result is correct. If any incorrect information is entered, the application will display an error message and allow the user to modify the answer and check it again.

Final result

Final remainder

0

¿Is it divisible?

☒ Yes

☐ No

Check final result

✓ Very good!

Final remainder: 0

The number 895958 is divisible by 7

Figure 6. Final result of the activity.

At the end of the activity, if you want to check another number using the same modulus, click the «New number» button. If you want to start a completely new activity with a different modulus and number, click the «New exercise» button.

Didactic value of the activity

This activity makes it possible to study divisibility in a visual and interactive way. Rather than simply obtaining the result of a division directly, students must actively participate in the entire process. The application allows them to observe how each digit of the number produces a transition between different states of the graph. In this way, students can relate the calculation of the remainder to modular arithmetic and understand that divisibility can also be represented by means of a state graph. The activity therefore combines mathematical reasoning with graphical representation, making the process of checking divisibility more visual, participatory, and easier to understand.